In this paper, we review in typical spaces in the general topology, from the Language of functions.

[Def.
Given a set X and a family of Topological space $\{Y_i\}_{i\in I}$ with functions

$$\{f_i : X \to Y_i\},$$

the topology $\mathcal{L}$ on X is called the initial topology if
$(X, \mathcal{L})$ is the coarest Topology such that for all $i\in I$, f_i is continuous

Equivalently,

$$\mathcal{L} = \bigcap \{\mathcal{L}' \mid \text{for all } i\in I,\ f_i : (X, \mathcal{L}') \to Y_i \text{ is continuous}\},$$

and generated by the subbasis $S := \{f_i^{-1}[U] \mid \text{for some } i\in I,\ U\subseteq Y_i \text{ open}\}$.
Dually, a set X and a family of Topological space $\{Y_i\}_{i\in I}$ with functions

$$\{g_i : Y_i \to X\}$$

the topology $\mathcal{L}$ on X is called the final topology if
$(X, \mathcal{L})$ is the finest Topology such that for all $i\in I$, g_i is continuous

Equivalently, the $\mathcal{L}$ is final topology if and only if $\mathcal{L}$ satisfies
$U\subseteq X$ is open set if and only if for each $i\in I$, $g_i^{-1}[U]\subseteq Y_i$ is open.

Moreover, given topological space Z and a map $h: Z\to X$, and $W: X\to Z$,
$h: Z\to (X, \mathcal{L}_{init})$ is continuous if and only if $f_i\circ h: Z\to Y_i$ is continuous, for all $i\in I$
$W: (X, \mathcal{L}_{final})\to Z$ is continuous if and only if $W\circ g_i: Y_i\to Z$ is continuous for all $i\in I$.

Diagramically,

$$Z \underset{W}{\overset{h}{\rightleftarrows}} X \underset{g_i}{\overset{f_i}{\rightleftarrows}} Y_i$$

Product Space

Tsub space

[Metric space

Let X be a set with a metric $d: X\times X \to \mathbb{R}$.

The Topology $\mathcal{T}_d$ generated by a basis $\{B_r(x) \mid x\in X, r>0\}$

is the coarset topology such that the metric d is continuous.

Prof. Given a basis element $(a,b)\subset\mathbb{R}$, $d^{-1}[(a,b)] = \{(x,y)\in X\times X \mid d(x,y)\in(a,b)\}$.

If $b\le 0$, then the preimage by d is empty, so open. Now, assume that $b>0$.

Let $(x,y)\in d^{-1}[(a,b)]$ be given. Then, $a< d(x,y)<b$,

take $r_x, r_y > 0$ such that $r_x + r_y < \min\{d(x,y)-a,\ b-d(x,y)\}$.

(It is possible, rigorously, take $r_x = r_y = \frac{1}{4}\cdot\min\{d(x,y)-a,\ b-d(x,y)\}$.).

Then, given $(x',y')\in B_{r_x}(x)\times B_{r_y}(y)$, that is, $d(x,x')<r_x$ and $d(y,y')<r_y$,

By the triangle inequality gives that

$$d(x',y') \le d(x',x)+d(x,y)+d(y,y') < r_x + d(x,y) + r_y < b$$

and

$$d(x,y) \le d(x,x')+d(x',y')+d(y',y) < r_x + d(x',y') + r_y$$

$$\Rightarrow a < d(x,y) - (r_x+r_y) < d(x',y'), \text{ so that}$$

$$a < d(x',y') < b, \text{ it means } (x',y')\in d^{-1}[(a,b)].$$

(We also used the symmetricity of metric), therefore

$$(x,y)\in B_{r_x}(x)\times B_{r_y}(y) \subseteq d^{-1}[(a,b)].$$

Since $B_{r_x}(x)\times B_{r_y}(y)$ is open set of $X\times X$, so $d^{-1}[(a,b)]$ is open. $\therefore d$ is continuous.

To show that the second assertion, let $\mathcal{T}'$ be a Topology on X such that the metric $d: X\times X\to\mathbb{R}$ is continuous. Let $x_0\in X$ be fixed.

Since every continuous map is continuous in each variable separately, the map $f_{x_0}: X\to\mathbb{R}; x\mapsto d(x,x_0)$ is continuous. We want to show that: $\mathcal{T}_d \subseteq \mathcal{T}'$.

Let $B_r(x)\in\mathcal{T}_d$ be an open set of the metric space. Then, for f_x,

$$B_r(x) = f_x^{-1}[(-\infty, r)]$$

is open set of $\mathcal{T}'$, since f_x is continuous on $\mathcal{T}'$ and $(-\infty,r)\subseteq\mathbb{R}$ is open set. ∎

the initial topology with respect to $\{ d_x : X \to \mathbb{R} : y \mapsto d(x,y) \mid x \in X \}$.